Table Of Content

Chapter 1: Introduction 1

Chapter 2: Problem Statement 2

Chapter 3: Objectives 3

Chapter 4: Methodology 4

4.1 Requirement Identification 4

4.1.1 Literature Review 4

4.1.2 Requirement Collection 5

4.2 Feasibility Study 7

4.2.1 Technical Feasibility 7

4.2.2 Operational Feasibility 7

4.2.3 Economical Feasibility 7

4.3 High-Level Design of Proposed System 8

Chapter 5: Expected Outcome 10

Chapter 6: References 11

List Of Figures

Figure 1 Use Case Diagram 6

Figure 2 Gantt Chart 8

Figure 3 System FlowChart 9

Chapter 1: Introduction

StegoCipher is a web-based secure messaging system that implements a hybrid cryptographic approach combining symmetric encryption, asymmetric encryption, and frequency-domain steganography. The system addresses the need for confidential communication by encrypting message content and concealing encryption keys within digital images using Discrete Wavelet Transform (DWT) techniques. Unlike conventional messaging applications that rely solely on end-to-end encryption, StegoCipher adds an additional layer of security by distributing decryption keys through separate, covert channels specifically, by embedding them imperceptibly within cover images. [1] This dual-layer approach ensures that even if encrypted messages are intercepted from the database, they remain unreadable without the corresponding key hidden in the carrier image. The system is designed for educational purposes to demonstrate fundamental concepts in cryptography, steganography, and secure system design. [2] It employs industry-standard algorithms including AES for symmetric encryption, RSA for asymmetric encryption, [3] and Argon2 for password hashing,[6] while utilizing DWT for robust image-based data hiding. [4]

Chapter 2: Problem Statement

In the digital age, the confidentiality and integrity of electronic communications are paramount concerns. Traditional messaging systems face several security challenges. Additionally, key exchange in asymmetric cryptography can be challenging without a secure channel, and visible encrypted data may attract attention from adversaries. [3] This project addresses the need for a system that not only encrypts messages but also conceals the encryption keys in a stealthy manner to evade detection. The core problems include:

- Conventional encrypted messaging systems store both encrypted messages and encryption keys (or key exchange mechanisms) within the same infrastructure. If an attacker gains access to the server or database, they may obtain sufficient information to compromise communications.
- Secure key exchange remains one of the most challenging aspects of cryptographic systems

By integrating hybrid encryption with steganography, the system aims to mitigate these issues in a controlled environment.

Chapter 3: Objectives

The primary objective of this project is to develop a secure messaging application that demonstrates the integration of hybrid cryptography and steganography for confidential communication. [4] Specific objectives include:

- Design and implement a functional secure messaging system that demonstrates the practical application of cryptographic algorithms (AES, RSA) and steganographic techniques (DWT).
- Develop a dual-channel security architecture, Channel 1: Encrypted messages stored in MySQL database and Channel 2: Encryption keys hidden in images via steganography
- Implement DWT-based steganography to embed RSA-encrypted symmetric keys into images [5]

Chapter 4: Methodology

The methodology follows a structured approach to system development, starting from requirement gathering to high-level design. It incorporates iterative planning and leverages existing cryptographic libraries for implementation.

4.1 Requirement Identification

This phase involves identifying the functional and non-functional requirements through literature review and collection methods.

4.1.1 Literature Review

A comprehensive review of existing literature on hybrid cryptography, steganography, and their integration for secure messaging systems reveals a growing emphasis on multi-layered security approaches to address vulnerabilities in standalone techniques. Hybrid encryption, which combines symmetric algorithms (e.g., AES for efficient bulk data encryption) with asymmetric algorithms (e.g., RSA for secure key exchange), offers enhanced security without significantly amplifying computational drawbacks. [3] Symmetric methods excel in handling large data volumes quickly but suffer from key distribution risks, while asymmetric methods provide robust key protection but are slower; their combination strengthens information security for applications like secure data transmission. This is particularly relevant for messaging, where hybrid systems mitigate interception and tampering in environments like cloud computing or electronic commerce. [1]

Steganography complements cryptography by embedding encrypted data (ciphertext) into digital media such as images, audio, or text, making hidden information undetectable and reducing the visibility that invites attacks like brute force or cryptanalysis. [5] Standalone cryptography ensures confidentiality but leaves ciphertext exposed, while standalone steganography (e.g., LSB substitution) hides data without encryption, rendering it vulnerable to extraction or steganalysis. [4] A hybrid approach addresses these gaps, providing comprehensive protection against modern threats through layered concealment and encryption

and Then comes the discrete wavelet transform (DWT) is a multiresolution analysis tool with excellent characteristics in the time and frequency domains. The coding efficiency and the quality of image restoration with the DWT are higher than those with the traditional discrete cosine transform and LSB. [2]

In Wavelets transformation, the Wavelets are the mathematical functions, defined at a fixed interval. [2]This has an average value of zero which transforms data into different frequency components by representing each component with a resolution matching its scale.

Further advancements include frameworks using Generative Adversarial Networks (GAN) for image steganography, achieving high embedding capacity and resistance to detection in big data contexts.

Argon2 is a key derivation function that was selected as the winner of Password Hashing Competition. The Argon2 function uses a large, fixed-size memory region (often called the 'memory array' in documentation) to make brute-force attacks computationally expensive. [6]

- Argon2d maximizes resistance to GPU cracking attacks.
- Argon2i is optimized to resist side-channel attacks. It accesses the memory array in a password-independent order.
- Argon2id is a hybrid version. It follows the Argon2i approach for the first half pass over memory and the Argon2d approach for subsequent passes.

4.1.2 Requirement Collection

Requirements were collected through stakeholder analysis (e.g., users, developers) and use case modeling based on the project description.

Functional Requirements:

- User registration with key generation,
- Message composition/encryption,
- Steganography for key hiding,

- Email delivery,
- Authenticated users shall view list of received messages
- Message Decryption

Non-Functional Requirements:

- Security (confidentiality)
- Usability (simple UI),
- Performance (quick encryption/decryption)

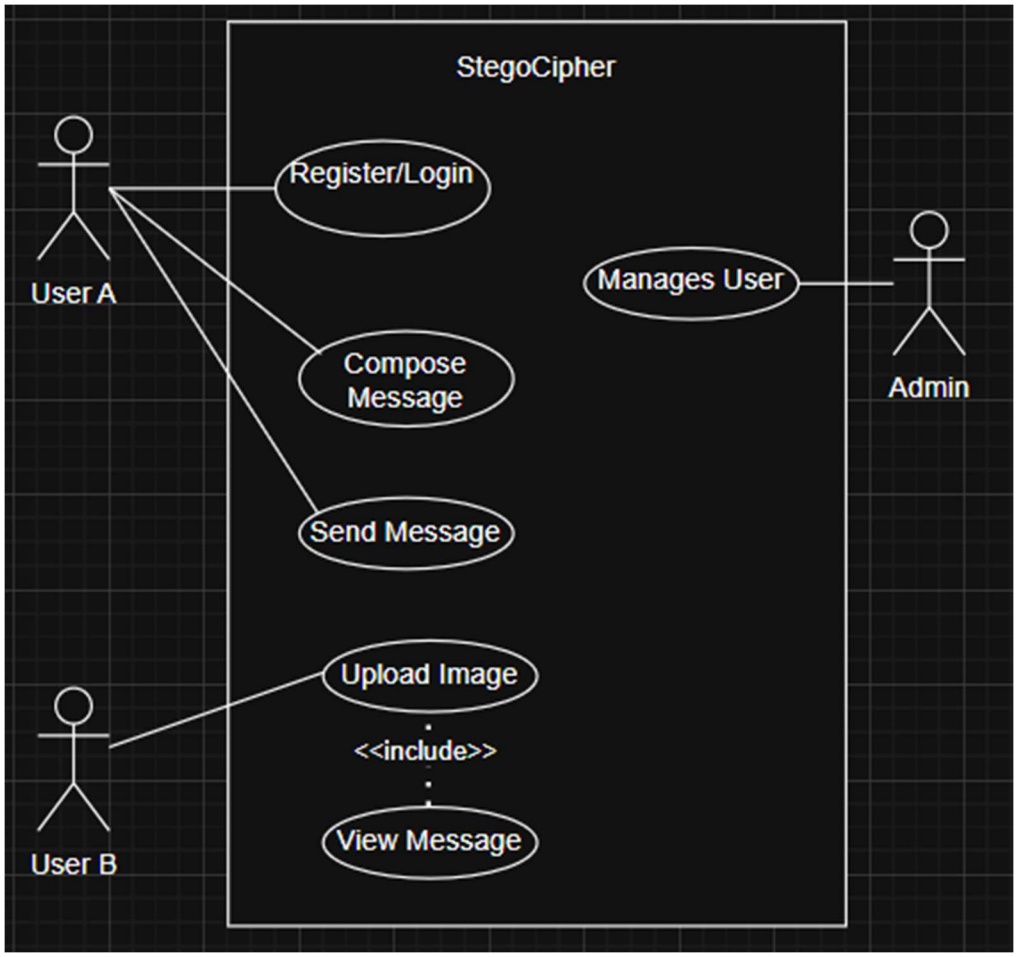

Figure 1 Use Case Diagram

4.2 Feasibility Study

This study assesses the practicality of the project across multiple dimensions.

4.2.1 Technical Feasibility

The project is technically feasible using open-source libraries:

- Database: MySQL for storing users, keys, and messages.
- Hardware requirements are minimal (standard PC with Python environment).
- No advanced infrastructure needed, as it's a local/web demo.

4.2.2 Operational Feasibility

Operationally, the system is user-friendly. Users need to register, compose messages, and upload images for decryption aligning with common app workflows. Training is minimal, but error handling (e.g., invalid keys) ensures robustness. In a real scenario, operational risks include email delays or spam filters.

4.2.3 Economical Feasibility

The project is cost-effective:

- Development Costs: Free tools (Python, open libraries); no licensing fees.
- Hardware/Software: Runs on personal computers.

The project timeline spans 10 weeks. Below is a tabular representation of the Gantt Chart:

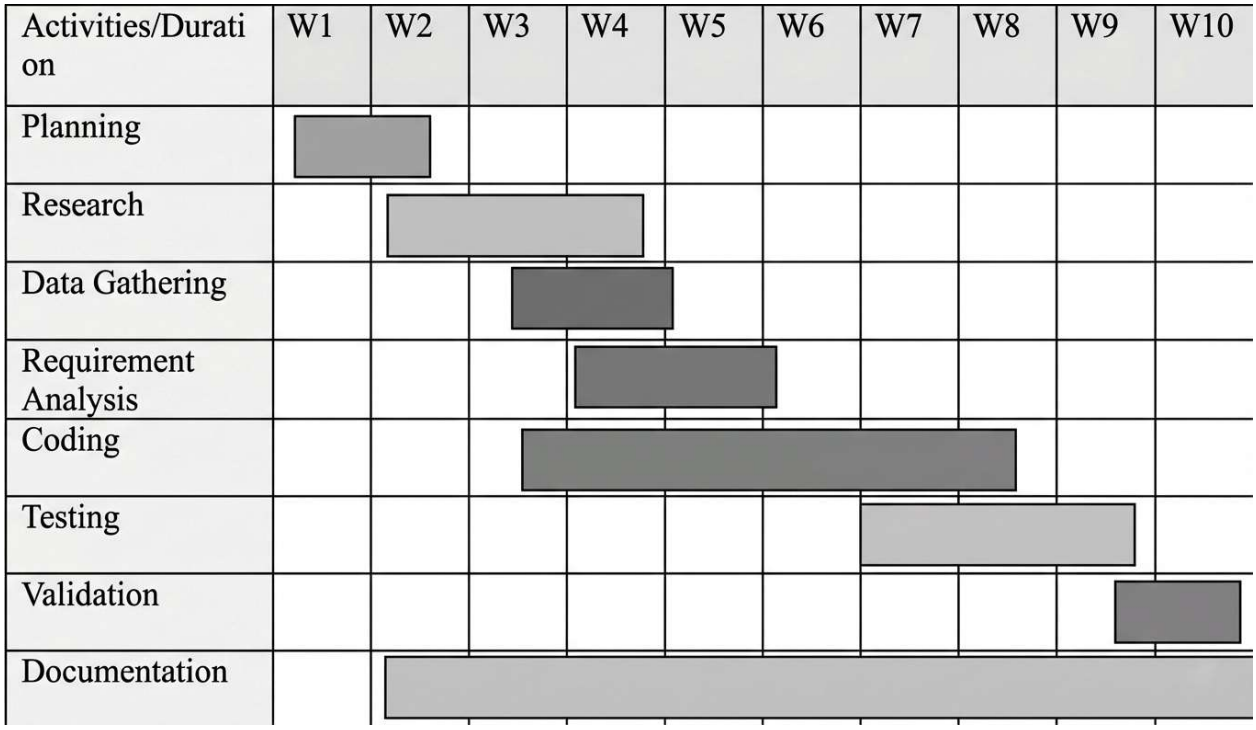

Figure 2 Gantt Chart

4.3 High-Level Design of Proposed System

The system architecture is modular:

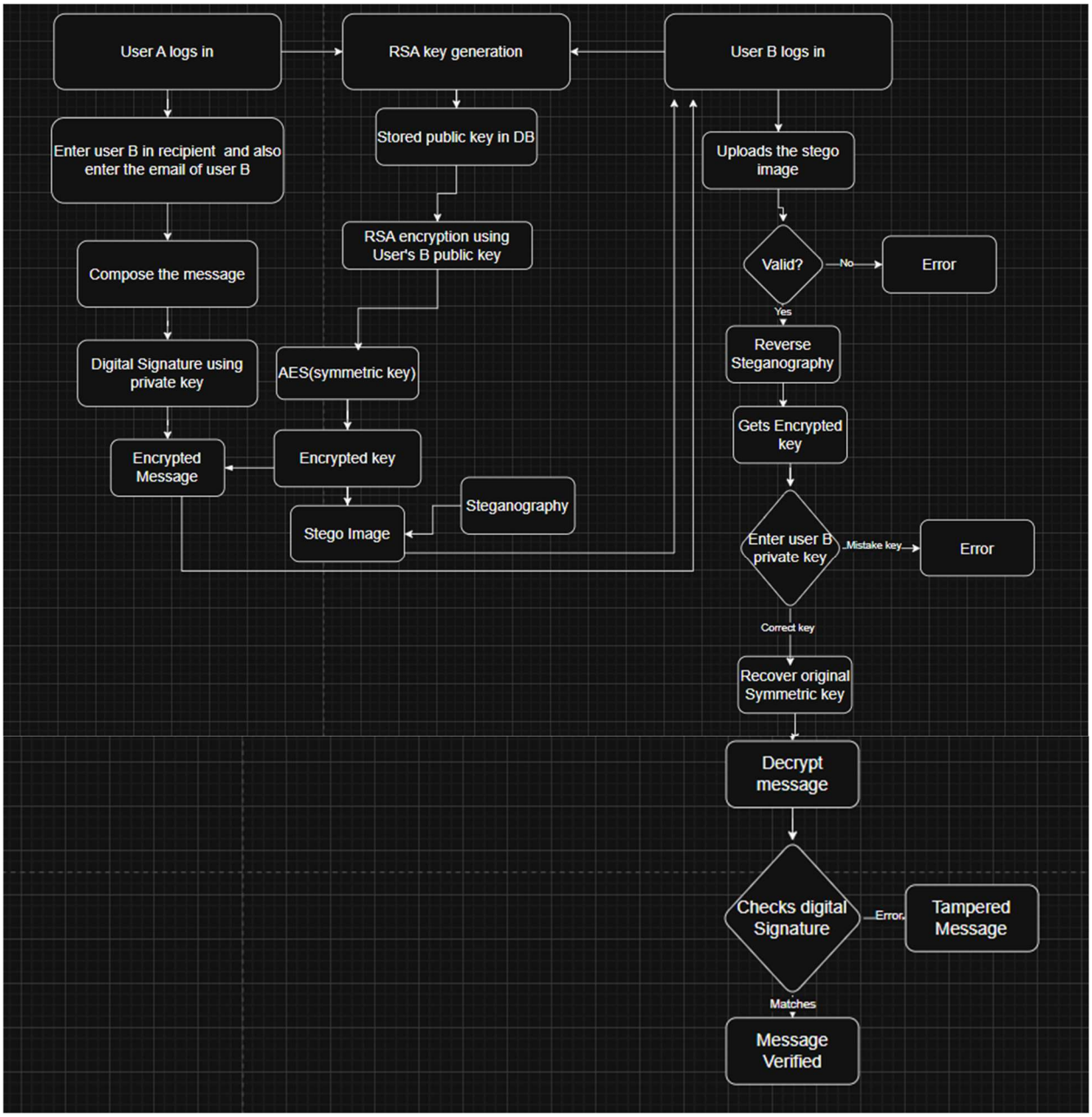

Figure 3 System FlowChart

Chapter 5: Expected Outcome

The expected outcome is a functional application that successfully registers users, sends encrypted messages with hidden keys via email, and allows decryption. It will demonstrate secure communication with no plaintext exposure during transit. Users will experience a closed, privacy-focused system including advanced stego techniques. [4] The project will validate the efficacy of hybrid approaches, achieving high integrity and confidentiality in tests.

Chapter 6: References

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